

AngularJS JSON

1. Create a table and retrieve the contents from JSON file, sort it using filters. Show the usage of \$index, \$odd and \$even.

Answer:

students.json:

```
[
  {
    "id":101,
    "name":"Suman",
    "course":"BCA",
    "marks":85
  },
  {
    "id":102,
    "name":"Rahul",
    "course":"BSc",
    "marks":78
  },
  {
    "id":103,
    "name":"Priya",
    "course":"BCom",
    "marks":92
  }
]
```

index.html:

```
<!DOCTYPE html>
<html ng-app="myApp">

<head>
<title>AngularJS JSON</title>

<style>
table{
  border-collapse:collapse;
  width:70%;
}
table,th,td{
  border:1px solid black;
  padding:10px;
}
.odd{
  background-color:#f2f2f2;
}
.even{
  background-color:#d9edf7;
}
</style>

</head>
```

```

<body ng-controller="myCtrl">

<h2>Student Details</h2>

<table>

<tr>
<th>Index</th>
<th>ID</th>
<th>Name</th>
<th>Course</th>
<th>Marks</th>
</tr>

<tr ng-repeat="student in students | orderBy:'marks'"
    ng-class="{ 'odd':$odd, 'even':$even}">

<td>{{ $index }}</td>
<td>{{ student.id }}</td>
<td>{{ student.name }}</td>
<td>{{ student.course }}</td>
<td>{{ student.marks }}</td>

</tr>

</table>

<script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"><
/script>

<script>

var app=angular.module("myApp", []);

app.controller("myCtrl", function($scope, $http) {

$http.get("students.json")

.then(function(response) {

$scope.students=response.data;

});

});

</script>

</body>
</html>

```

Output Features

- Reads data from students.json.
- Displays data in a table.
- Sorts records using orderBy.
- Uses:
 - **\$index** -> Displays row number.
 - **\$odd** -> Styles odd rows.
 - **\$even** -> Styles even rows.

2. Create a select element with some options. Spot the difference between ng-options and ng-repeat by using both inside the program.

Answer:

```
<!DOCTYPE html>
<html ng-app="myApp">

<head>
<title>ng-options vs ng-repeat</title>
</head>

<body ng-controller="myCtrl">

<h3>Using ng-options</h3>

<select ng-model="selected1"
        ng-options="city for city in cities">
</select>

<p>Selected: {{selected1}}</p>

<br>

<h3>Using ng-repeat</h3>

<select ng-model="selected2">

<option ng-repeat="city in cities">
{{city}}
</option>

</select>

<p>Selected: {{selected2}}</p>

<script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"><
/script>

<script>

var app=angular.module("myApp", []);

app.controller("myCtrl", function($scope) {
```

```

$scope.cities=[
  "Bangalore",
  "Hyderabad",
  "Chennai",
  "Mumbai"
];

});

</script>

</body>
</html>

```

Difference between ng-options and ng-repeat:

ng-options	ng-repeat
Designed specifically for <select> elements	General-purpose directive for repeating HTML elements
Faster and more efficient for large lists	Slower for large lists because it creates many DOM elements
Automatically binds objects and values	Repeats <option> elements manually
Recommended for dropdown lists	Used when custom option formatting is required

Things to know:

1. Understand the concept of JSON?

What is JSON?

JSON (JavaScript Object Notation) is a lightweight data format used to **store and exchange data** between a client and a server.

Features

- Lightweight and easy to read.
- Human-readable format.
- Language-independent.
- Commonly used with REST APIs and AngularJS.

JSON Example

```

{
  "id":101,
  "name":"Suman",
  "course":"BCA",
  "marks":85
}

```

JSON Array Example

```

[
  {
    "id":101,
    "name":"Suman"
  },
  {
    "id":102,

```

```
        "name": "Rahul"
    }
]
```

Advantages of JSON

1. Easy to read and write.
2. Lightweight compared to XML.
3. Fast data exchange between client and server.
4. Supported by almost all programming languages.
5. Widely used in web services and REST APIs.